

ALOX "Primed Instability" Immune Interface: Lipid Peroxide–Driven Immunosuppression

NRF2–High OSCC co–upregulates pro–ferroptotic lipoxygenases (ALOX12/12B/15) alongside antioxidants — creating a high lipid peroxide state.
ALOX15–derived 15–HETE and ALOX12 products suppress NK cytotoxicity and promote M2 polarisation via PGE2–like immunomodulation.
Novel: simultaneous 3–ALOX upregulation not previously characterised in OSCC.

